16.15. $\Delta\delta = 2\times(n-1)\times 100\,mm$

$= 100\times 589\,nm.$

$$n = \frac{100\times 589\times 10^{-9}\,m}{100\times 10^{-3}\,m\times 2} + 1 = 1.0002945$$

16.19. $2ne < \frac{\lambda^2}{\Delta\lambda}$

$$e < \frac{(500\,nm)^2}{0.05\,nm\times 2\times 1.3} = 1.92\,mm$$

17.1 $\Delta x = 2f\tan\theta$

$$\theta = \frac{\lambda}{a}$$

$$\Delta x = 2f\tan\theta = 2f\cdot\frac{\lambda}{a} = \frac{2\times 60\,cm\times 632.8\,nm}{0.2\,mm} = 3.7968\,mm$$

17.3. $\Delta d = \frac{\Delta x}{2} = f\tan\theta = 50\,cm\cdot\frac{\lambda}{0.5\,mm}\times(1+2n)$

$$\lambda = \frac{1.5\,mm\times 0.5\,mm}{50\,cm\,(1+2n)} = \frac{1500}{(1+2n)}\times 10^{-9}\,m.$$

$n=1$ 时，$\lambda = 500\,nm$，是绿光

17.5. $\frac{2\lambda}{a} = \sin\varphi \pm \sin\theta.$

$$a = \frac{2\lambda}{\sin\varphi\pm\sin\theta} = \frac{1000\,nm}{\frac{1}{2}\pm 5.0039}$$

同侧，$a = \frac{1000\,nm}{0.0039} = 0.26\,mm$

异侧，$a = \frac{1000\,nm}{1.0039} = 1\,\mu m.$

(1)

17.6. $-\frac{\lambda}{a} \le \sin\alpha \le \frac{\lambda}{a}$, $-6\times10^{-3} \le \sin\alpha \le 6\times10^{-3}$

$d\sin\theta = \pm k\lambda$. $\sin\theta \le \sin\alpha$.

$\sin\theta = \pm k \cdot \frac{480nm}{0.4mm} = \pm 1.2\times10^{-3} \cdot k$.

$k = 0, 1, 2, 3, 4, 5$.

则干涉亮纹共9条.

(2). $\Delta x = f\frac{\lambda}{d} = 2.4mm$

17.11 (1). $\sin\theta \cdot d = \lambda$.

$d = \frac{\lambda}{\sin\theta} = \frac{632.8nm}{\sin 38^\circ} = 1.03\times10^{-3}mm$.

(2). $\lambda = d\sin\theta$

$= 1.03\times10^{-3}mm \times \sin 27^\circ$

$= 467.6nm$.

$k < \frac{d}{\lambda} = \frac{1}{\sin 27^\circ} = 2.2$.

则 $k_{max} = 2$.

最多可看到第2级明条纹.

17.13

光栅方程: $d(\sin i \pm \sin\theta) = \pm k\lambda$, $k \in \mathbb{Z}$.

单缝衍射: $a(\sin i \pm \sin\theta) = \pm k'\lambda$, $k' \in \mathbb{Z}$.

$\frac{a}{d} = \frac{k'}{k} = \frac{2}{3}$, $k = \frac{3}{2}k'$

则 $k = 3n$, $n = 1, 2, 3, \ldots$ 时, 会缺级.